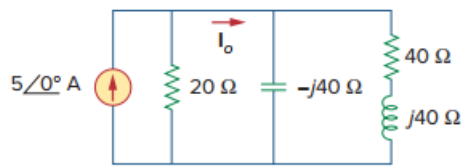


Problem

Find current I_o in the circuit shown in Fig. 9.50.



Step-by-step solution

Step 1 of 2

Refer to Figure 9.50 in the text book for the circuit.

Determine the equivalent impedance Z_{eq1} of the parallel combination of capacitance, and resistance and inductance in series to the right side of the circuit.

$$\begin{aligned} Z_{eq1} &= (-j40) \parallel (40 + j40) \\ &= \frac{(-j40)(40 + j40)}{40 + j40 - j40} \\ &= 40 - j40\ \Omega \end{aligned}$$

Step 2 of 2

Apply current division rule to find the current, I_o .

$$\begin{aligned} I_o &= (5\angle 0^\circ) \left(\frac{20}{20 + Z_{eq1}} \right) \\ &= (5\angle 0^\circ) \left(\frac{20}{20 + 40 - j40} \right) \\ &= \frac{15}{13} + j\frac{10}{13} \\ &= 1.3868\angle 33.69^\circ\text{ A} \end{aligned}$$

Thus, the current I_o in the circuit is $1.3868\angle 33.69^\circ\text{ A}$.